INDEPENDENT WATCHDOG AND EDGE–ARMED GLOBAL PERMISSION – ENGINEERING CANDIDATE

Seven ICs; no firmware timer owns the external watchdog. IO_3V3=3.2–3.4V, common LOGIC_GND. Eight motors minimum / twelve expansion requirement.

120pF:64.288ms nominal watchdog. NOT20ms.
100pF nominal would violate tested minimum at –tolerance.
Falling progress edge occurs every TWO accepted UPDATEs.
WDO auto–releases after170–230ms; heartbeat alone
cannot set the arm latch. SYNC + deliberate ARM required.
CDONE pullup belongs to FPGA block: no local pulldown.

HW_WATCHDOG_OK_N –> RTL inhibit_n. Never return TX_PERMIT there.
Raw interlock has no endpoint_ready / motion_request / latch dependency.
D is permanently HIGH; clock is FPGA_MOTION_REQUEST rising edge.
Asynchronous clear is raw interlock AND endpoint_ready.
TX_PERMIT = latched_arm AND motion_request AND raw_interlock.

TX_PERMIT drives six AM26C31 G inputs + six10k pulldowns; /G fixedHIGH.
DefaultLOW disables transmitters. Per–drive active–low enables are separate.
EXT_PERMIT_3V3 is protected internal logic, never raw24V or certified E–stop.
Single–channel candidate: no safety certification, fault tolerance or controlled stop.
Brownout slew, partial power, metastability, wiring and physical timing remain open.
Clearing output permission can truncate a pulse; reconcile position before recovery.

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